

TITANIUM-DIFFUSED LITHIUM NIOBATE WAVEGUIDE DEVICES

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Abstract

The current status of devices and technology based upon waveguides formed by titanium diffusion into lithium niobate is reviewed.

1. Introduction

In the slightly more than ten years since the first demonstration of optical waveguides [1] formed in lithium niobate by thermal indiffusion, devices based on this technology have move from research laboratory magic shows to recently announced commercial products. While there is much yet to be done - optimize device performance, increase the level of integration, demonstrate mass manufacturability and reliability - titanium diffused lithium niobate devices clearly will play an important role in the rapidly emerging lightwave network and probably in signal processing and sensing applications as well. In this talk we will review the titanium-diffusion process technology, describe a few of the large number the diverse devices that have been demonstrated using this technology, describe some exciting applications and point out some remaining material questions [2].

2. Fabrication

The process of fabricating titanium diffused lithium niobate waveguides is shown in Fig. 1. The desired waveguide pattern is photolithographically defined on the substrate.

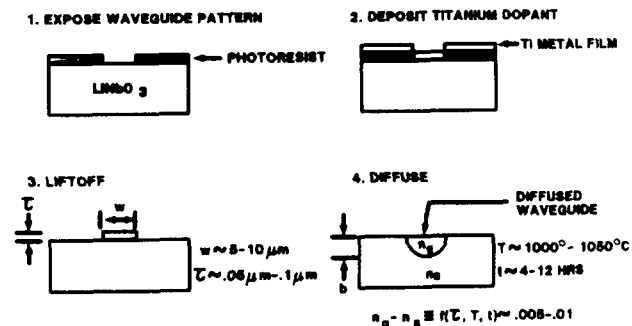
Titanium is deposited, typically by a resistively heated evaporator, E-beam or sputtering, over the entire substrate. By lift-off the titanium is removed from the areas outside the patterned region leaving a metalized pattern corresponding to the desired waveguide pattern. Typical strip widths are from $3\text{ }\mu\text{m}$ for visible operation to $8\text{ to }10\text{ }\mu\text{m}$ for operation at wavelengths of $1.3\text{ }\mu\text{m}$ to $1.5\text{ }\mu\text{m}$ where optical fiber has its lowest dispersion and loss, respectively. Metal thickness values vary from about $0.3\text{ }\mu\text{m}$ to $1.00\text{ }\mu\text{m}$, depending upon wavelength and the degree of confinement desired. The metal is then diffused into the crystal at temperatures between 980°C and 1050°C for times in the 4 to 12 hour range. The diffusion atmosphere- flow gases, containment vessel, etc- varies greatly from one laboratory to another. These various techniques are used to eliminate unwanted lithium out-diffusion which typically accompanies the in-diffusion process and can lead to undesirable surface waveguides outside the channel region. While the detailed mechanism of these techniques are not fully understood several methods, including performing the diffusion in a water vapor atmosphere, produce waveguides for which surface guiding is not a problem. After diffusion the crystal ends are carefully polished to allow end-fire coupling either with a lens or by butt coupling with fiber.

An important advantage of this technology, in addition to being able to take advantage of the excellent electro-optic, acousto-optic and transmission characteristics and commercial availability of lithium niobate, is that the diffusion process allows good control over the waveguide mode size. This provides the flexibility to achieve excellent butt-coupling to single-mode optical fibers where matching the mode size of the fiber and waveguide is essential. Indeed, fiber to waveguide coupling losses as low as .3 dB have been demonstrated. For the same diffusion parameters, waveguide propagation losses are .3 dB/cm for wavelengths in the $1.3\text{ to }1.5\text{ }\mu\text{m}$ range. Low fiber-coupling losses are, of course critical for applications in lightwave communication systems.

3. Devices

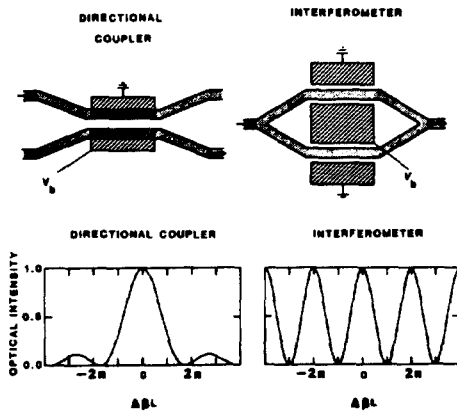
A large variety of devices have been demonstrated using the Ti:LiNbO_3 technology. Examples include switch/modulators, acousto-optic deflectors, tunable wavelength filters, polarization controllers and splitters, frequency shifters and doublers. Two of the most important waveguide devices are the Y-branch interferometric modulator and the directional coupler switch, shown schematically in Fig. 2.

Ti-DIFFUSED LITHIUM NIOBATE WAVEGUIDES



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ELECTRO-OPTIC WAVEGUIDE MODULATORS



In the Y-branch interferometer light in the input single mode waveguide is equally split by the Y-branch into the two arms of the interferometer, light propagates down the isolated arms where the relative phase is changed via the electro-optic effect. The two waves are combined by the output Y-branch. If a π phase shift is introduced between the two beams, the combined mode is double lobed and is not matched to the single-mode output waveguide. As a result light is radiated into the the substrate and the output level is low. For no induced phase shift between the two arms, the combined beam is identical to that input and the output is high. The optical transmission versus applied voltage ($\Delta\beta$ in Fig. 2 is proportional to the applied voltage) is sinusoidal as shown in Fig. 2.

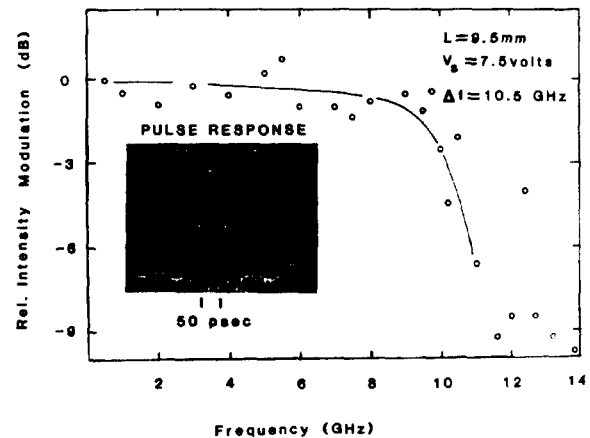
The directional coupler switch consists of a pair of single-mode channel waveguides placed in close proximity over some interaction length. Coupling between the two waveguides occurs because the evanescent tail of the optical mode of one waveguide extends into the other. As a result, light input into one waveguide couples into the other as it propagates down the waveguide. Indeed, if the speed of propagation of light in the two waveguides is identical, i.e. they have the same refractive indices, light is entirely transferred to the second waveguide after a characteristic length referred to as the coupling length. The coupling length depends upon the mode confinement, wavelength and interwaveguide separation. However, if the propagation speed in the two waveguides is sufficiently different, there is no coherent addition of the small coupling contributions along the interaction length and the net coupling is small. As a result, light exits from one or the other output waveguides depending upon the applied voltage. The cross-over or transfer efficiency versus applied voltage is also shown in Fig. 2. Note that the ratio of light in the two waveguides can be continuously varied so the device can also be used as a voltage controlled tap.

For both the interferometer and the directional coupler the electro-optically induced phase shift ($\Delta\beta L$) to achieve complete modulation is approximately π . Because of the small lateral dimensions and the ability to use a rather long interaction length, the required switch voltage can be much lower than for bulk devices. A useful practical figure of merit of the switch/modulator voltage is the voltage $\times$ length product. For Ti:LiNbO_3 devices that utilize the strong r_{33} coefficient, this number is 1 volt-cm for operation at a wavelength of $0.6 \mu\text{m}$ and 5 volt-cm for $1.3 \mu\text{m}$.

4. Switching Speed

Very high-speed Ti:LiNbO_3 waveguide switch/modulators have been demonstrated. The response time of the electro-optic effect in lithium niobate is, of course, very fast, < 1 picosecond. However, the actual switching speed is limited by circuit effects. Best results have been achieved using a traveling wave electrode in which the electrical drive signal propagates on a coplanar transmission line placed over or along the waveguide. The response time or bandwidth in this case is not limited by capacitive charging effects but rather by the difference in velocity between the co-propagating electrical and optical fields. In lithium niobate this sets an inherent bandwidth $\times$ length limitation of $10 \text{ GHz}\cdot\text{cm}$. Measured frequency and pulse response results for a 9.5 mm long directional coupler switch operating at a wavelength of $1.3 \mu\text{m}$ is shown in Fig. 3. At shorter wavelengths where modulation at similar voltages can be achieved, modulation bandwidths as large as 17 GHz have been achieved.

FREQUENCY RESPONSE



5. Level of Integration

In spite of the relatively good electro-optic characteristics of lithium niobate, a device length of several millimeters is necessary for drive voltages that can be achieved at high modulation rates. Lateral dimensions on the chip, however, are much smaller. Therefore, the largest integration has been achieved for arrays of identical devices with several parallel inputs. The currently most important example is crossbar switch arrays. A schematic drawing of a 4×4 directional coupler array in which any of four inputs can be switched to any of four output ports is shown in Fig. 4. Such arrays, which integrate sixteen couplers have been demonstrated with an insertion loss of 6 dB and crosstalk isolation better than 30 dB [3]. Very recently a 8×8 switch array that integrates 64 directional couplers on a 6 cm long crystal has been reported [4]. These switch arrays are being pursued for applications in video switching and for switched wideband services in future lightwave loop networks.

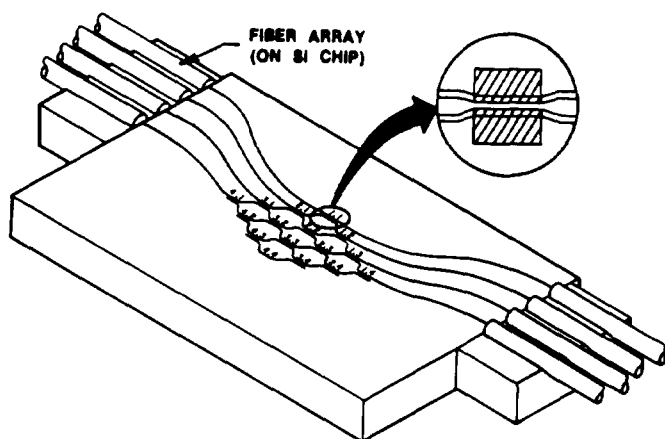


DIAGRAM OF A 4 X 4 DIRECTIONAL COUPLER SWITCH MATRIX

In addition, chips that integrate several different waveguide devices on a chip to make a photonic circuit to provide a larger system function have been demonstrated. Examples include the RF spectrum analyzer, polarization controller and a coherent lightwave receiver that combines a directional coupler, polarization controller and frequency shifter on a single chip.

6. Applications

Titanium diffused lithium niobate devices will eventually find application in communication, signal processing and sensor systems. We highlight two examples here. Modulators appear particularly attractive for high-data-rate, long distance lightwave systems. This is particularly true for coherent systems where phase modulation is not easily achievable directly by the laser [5]. In fact, the best lightwave system performance to date (highest information rate over the longest unrepeated distance) has been achieved with a broadband, low loss traveling-wave Ti:LiNbO_3 directional coupler modulator. Information at a data rate of 8 Gigabits/sec has been transmitted over 68 kilometers using encoding by external modulation [6]. System researchers around the world now routinely use these devices [7]. A photograph of a packaged, fiber pigtailed device used in lightwave transmission experiment system experiments is shown in Fig. 5. As the sophistication of lightwave systems increases, there will be need for increased functionality which can be provided by the large bag of tricks that lithium niobate devices offer.



In another important application area, a multi-function chip that provides the splitting, phase modulation and combining functions has been developed for a fiber gyroscope. In this application the potential for low cost, compactness and ability for electro-optic control provides potentially important advantages over other approaches.

7. Conclusions

Titanium diffused lithium niobate device technology is clearly a maturing technology with important identified applications. Some material work remains to be done and questions of mass manufacturability remain to be addressed. In addition, it will be important to identify applications with sufficient volume to make the mass production essential to economically justify complete development of the technology. Nevertheless, the prospect that this technology will make a significant impact on optical systems of the future is, I believe, very good.

References

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